

# MazeGame

## Development Handoff Document

Unity 6 | Built-In Render Pipeline | Legacy Input System

### 1. Folder Structure

All game assets live under the Assets/ directory. Below is a complete breakdown of every folder, what it contains, and its current status.

#### 1.1 Top-Level Overview

| Folder       | Contents                          | Notes                                                            |
|--------------|-----------------------------------|------------------------------------------------------------------|
| Animations/  | Character animation clips         | Currently empty — animations live inside ThirdParty asset folder |
| Audio/Music/ | Background music tracks           | Empty — not yet implemented                                      |
| Audio/SFX/   | Sound effects                     | Empty — key pickup, death, win, lose, evil laugh not added yet   |
| Materials/   | Unity materials for game objects  | 4 materials created (see below)                                  |
| Prefabs/     | Reusable game object prefabs      | Keys and Player ready; Obstacles empty                           |
| Scenes/      | All Unity scenes                  | 4 scenes: MainMenu, LevelSelect, MazeGame (old), Maze1 (active)  |
| Scripts/     | All C# game scripts               | Organized into subfolders by category                            |
| TextMeshPro/ | Auto-generated by Unity           | Do NOT modify or delete                                          |
| ThirdParty/  | All imported Asset Store packages | 3 packages imported                                              |
| UI/          | Canva-created images for UI       | Buttons and background images organized by scene                 |

#### 1.2 Materials Folder

| Folder        | Contents                                  | Notes                          |
|---------------|-------------------------------------------|--------------------------------|
| KeyMat_Yellow | Shiny yellow material applied to all keys | Metallic: 0.8, Smoothness: 0.9 |

|            |                                                 |                                                   |
|------------|-------------------------------------------------|---------------------------------------------------|
| KeySparkle | Star Pack particle material for key glow effect | Uses Andtech Star Pack shader                     |
| WallMat    | Green wall material (original)                  | Not currently in use — walls use asset materials  |
| FloorMat   | Floor/grass material (original)                 | Not currently in use — floor uses asset materials |

### 1.3 Prefabs Folder

| Folder             | Contents                                                       | Notes                                                        |
|--------------------|----------------------------------------------------------------|--------------------------------------------------------------|
| Prefabs/Keys/      | Key FBX mesh + Box Collider (Is Trigger) + KeySparkle particle | KeyPickup.cs attached to each key instance                   |
| Prefabs/Player/    | FirstPersonController prefab from Modular FPS package          | Capsule Collider + Rigidbody + FPS script                    |
| Prefabs/Obstacles/ | Empty — obstacles not built yet                                | Will contain pop-up obstacle prefabs for Level 2 & 3         |
| Prefabs/Maze/      | Entire Maze1 layout (walls + floor) as a single object         | Imported from asset pack; collision handled by Mesh Collider |

### 1.4 Scenes Folder

| Folder      | Contents                                 | Notes                                           |
|-------------|------------------------------------------|-------------------------------------------------|
| MainMenu    | Start button, Levels button, Quit button | Active and working                              |
| LevelSelect | Level 1, Level 2, Level 3, Back buttons  | Active; Level 2 & 3 load scenes not yet created |
| MazeGame    | Old prototype maze scene                 | DO NOT USE — kept as backup only                |
| Maze1       | Main gameplay scene — Level 1            | Active and working                              |

### 1.5 Scripts Folder

Scripts are organized into subfolders by responsibility:

| Script          | Location          | What It Does                                                                                |
|-----------------|-------------------|---------------------------------------------------------------------------------------------|
| GameManager.cs  | Scripts/Managers/ | Core game logic: timer countdown, key tracking, win/lose/exit conditions, scene transitions |
| SpawnManager.cs | Scripts/Managers/ | Places player at correct spawn point when scene loads                                       |

|                       |                     |                                                                                         |
|-----------------------|---------------------|-----------------------------------------------------------------------------------------|
| MainMenu.cs           | Scripts/UI/         | Handles Start, Levels, and Quit button click events on MainMenu scene                   |
| LevelSelectManager.cs | Scripts/UI/         | Handles Level 1/2/3 and Back button click events on LevelSelect scene                   |
| FloatingEffect.cs     | Scripts/UI/         | Applies a gentle up-and-down floating animation to objects (used on keys)               |
| KeyPickup.cs          | Scripts/Player/     | Detects when Player1 enters key trigger collider; notifies GameManager and destroys key |
| ExitTrigger.cs        | Scripts/Game State/ | Detects when Player1 reaches the exit; stops movement and triggers win                  |

## 1.6 ThirdParty Folder

| Folder                        | Contents                                        | Notes                                                         |
|-------------------------------|-------------------------------------------------|---------------------------------------------------------------|
| ModularFirstPersonController/ | Full FPS controller with camera, head bob, jump | Used as Player1 prefab; WASD/Arrow keys; mouse look           |
| Andtech/StarPack/             | Particle sparkle shader and textures            | Used for glowing key particle effect                          |
| HalloweenPack/                | Modular cartoon character + animations + props  | Halloween-themed assets; used for maze environment decoration |
| Bushes/                       | Bush FBX mesh + leaf textures from OpenGameArt  | Imported but not currently used in active scene               |

## 1.7 UI Folder

| Folder       | Contents                                               | Notes                                                     |
|--------------|--------------------------------------------------------|-----------------------------------------------------------|
| UI/Buttons/  | Button images exported from Canva (PNG, Sprite format) | Start, Levels, Quit, Level 1/2/3 button images            |
| UI/MazeGame/ | Background and decorative images from Canva            | Main menu background (1920x1080), level select background |

## 2. What Has Been Completed

| Phase                   | Description                                         | Status |
|-------------------------|-----------------------------------------------------|--------|
| Phase 1 — Project Setup | Folders, tags, scenes, input axes, build profiles   | Done   |
| Phase 2 — Player & Maze | FPS controller, maze layout, materials, spawn point | Done   |

|                          |                                                                          |             |
|--------------------------|--------------------------------------------------------------------------|-------------|
| Phase 3 — Key Collection | Key prefabs, colliders, KeyPickup script, GameManager integration        | Done        |
| Phase 4 — Exit & Win     | Exit trigger, player freeze on win, win condition logic                  | Done        |
| Phase 5 — Timer          | 3-minute countdown in GameManager, lose on timeout                       | Done        |
| Phase 6 — Main Menu      | MainMenu scene with Start/Levels/Quit buttons and background             | Done        |
| Phase 7 — Level Select   | LevelSelect scene with Level 1/2/3 and Back buttons                      | Done        |
| Phase 8 — Win/Lose UI    | Win screen (Replay/Next Level/Main Menu), Lose screen (Replay/Main Menu) | In Progress |
| Phase 9 — Obstacles      | Pop-up obstacles for Level 2 & 3 with warning sounds                     | Pending     |
| Phase 10 — Audio         | Key pickup, death, win, lose, evil laugh SFX + background music          | Pending     |
| Phase 11 — Level 2 & 3   | Duplicate maze with adjusted key positions and obstacles                 | Pending     |
| Phase 12 — Polish        | Visual improvements, particle effects, final testing                     | Pending     |

## 3. What Is Still Left To Do

### 3.1 Win & Lose Screens

- Win Screen needs: title text, Replay button, Next Level button, Main Menu button
- Lose Screen needs: title text, Replay button, Main Menu button
- Both screens need visual styling to match the game's Halloween theme
- UIManager.cs needs to be created and wired to GameManager

### 3.2 HUD (In-Game UI)

- Timer display (TextMeshPro) — top center of screen
- Key counter display (e.g. 1/3) — top left of screen
- Exit Level button — top right of screen

- UIManager.cs needs to update these elements in real time (script not created yet)

### 3.3 Audio System

- Key pickup sound — plays when player collects a key
- Walking sound (optional if there's play music)
- Win sound — plays on win screen
- Lose/death sound — plays on lose screen
- Spawn sound — plays when player first spawns
- Evil laugh sound — plays when player clicks Exit Level button or loses //game taunting player for giving up/ losing
- Eerie Background music — ambient loop during gameplay
- AudioManager.cs script needs to be created in Scripts/Audio/

### 3.4 Levels 2 and 3

- Create Maze2 scene (duplicate Maze1 with modified key positions and obstacles)
- Create Maze3 scene (harder version with more keys and faster obstacles)
- Add obstacle prefabs to Prefabs/Obstacles/ //you may use any of the Halloween assets as obstacles
- Write obstacle spawning script in Scripts/Obstacles/
- Lives system (3 lives) for Level 2 and Level 3

### 3.5 Fixes Required

- Gameplay timer should be adjusted to 10 minutes (currently 3 minutes — too short)
- Level 1 pacing feels slow due to long walking distances between keys with no interactions
- Exit level button mid-gameplay needs evil laugh audio hooked up
- MazeGame scene (old) should be deleted or clearly marked to avoid confusion
- Add pointing hand animation in all levels (dana's idea: absolute cinema)

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## 4. Level Design Suggestions

### 4.1 Level 1 — Easy (Current State)

Level 1 is working as intended. It serves as a tutorial level with no obstacles.

- 3 yellow keys placed at visible, reachable locations
- No obstacles — pure exploration
- 10-minute timer gives plenty of time
- Exit unlocks only after all 3 keys collected

### 4.2 Level 2 — Medium (Recommended Design)

Two approaches are suggested — pick one or combine elements from both:

### Option A: Hidden Keys + Obstacles

- Keep same maze layout
- Move keys to harder-to-find locations (dead ends, corners, behind turns)
- Add 2 pop-up obstacles that appear suddenly in corridors
  - Warning sound plays 2 seconds before obstacle appears (optional?)
  - Collision with obstacle = lose 1 life (3 lives total)
- Player has 3 lives — shown in HUD

### Option B: Same Keys + Harder Obstacles

- Keep original key positions from Level 1
- Add more interactive elements in corridors to reduce empty walking feeling
  - Breakable crates (decorative, using HalloweenPack props)
  - Flickering lights near obstacle zones for atmosphere
- 2 obstacles that move slightly (patrol a short path)
- Shorter warning time before obstacle appears (1 second instead of 2)

*Recommendation: Use Option A for clearest difficulty progression. Add a few decorative props from HalloweenPack to break up long corridors and improve pacing.*

## 4.3 Level 3 — Hard (Recommended Design)

- 5 yellow keys instead of 3 — two extra keys in hard-to-reach locations
  - 15-minute timer
  - 3 obstacles instead of 2
  - Obstacles appear faster and more frequently
    - Warning sound plays only 0.5 seconds before appearing
    - Obstacles may appear in groups of 2 simultaneously
  - Player still has 3 lives
  - Consider adding a fog/darkness visual effect using Unity Post Processing
  - Keys could glow less visibly — reduce particle effect intensity
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## 5. Known Fixes & Issues

### 5.1 Timer Duration

Current timer is set to 3 minutes (180 seconds). This is too short for a maze game, especially for players unfamiliar with the layout.

- Fix: Change `levelTime = 180f` to `levelTime = 600f` in `GameManager.cs`
- This gives players 10 minutes which is much more comfortable
- Level 3 could optionally use 8 minutes to increase difficulty // nope bad idea, too hard

## 5.2 Level 1 Pacing

Level 1 currently feels boring because players walk long distances between keys with no environmental interactions or events to break the monotony.

- Fix: Add interactions to decorative props from HalloweenPack along key paths (pumpkins, gravestones, lanterns)
- Fix: Add ambient particle effects (falling leaves, floating dust) to make corridors feel alive
- Fix: Add a flickering light effect near obstacle zones using Unity's Light component animation

## 5.3 Other Known Issues

- MazeGame scene (old prototype) should be deleted from Scenes/ to avoid confusion
- Evil laugh sound effect is not yet hooked up to the Exit Level button
- UIManager.cs is referenced in GameManager but not yet created — causes compiler warning
- Level 2 and Level 3 scenes do not exist yet — LevelSelect buttons will fail silently